Introduction to Speech LLMs

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Ref: On The Landscape of Spoken Language Models: A Comprehensive Survey

Why we should learn?

ChatGPT OpenAI

Gemini Google

Claude ANTHROP\C

Perplexity AI perplexity

aws Amazon Titan aws

ERNIE Bot Bai du 百度

DeepSeek-V2 DeepSeek

Llama 3 Meta

Mistral Large MISTRAL AI_

cohere Command R+ cohere

Microsoft Copilot Microsoft

Grok (xAI) xAI

Qwen2 / Qwen2.5 Alibaba

Qwen-Audio Alibaba

SpeechGPT Microsoft

VALL-E 2 Microsoft

ElevenLabs ElevenLabs

Meta Voicebox Meta

Code Llama Meta

StarCoder 2 BigCode

Med-PaLM 2 Google

BioGPT Microsoft

FinGPT FinGPT

Why we should learn?

We can always do better!

- Paul Wellstone

Current Limitations of Speech LLMs:

1. Hallucinations and unreliable responses
2. Sensitivity to noise and ASR errors
3. Limited understanding of prosody (stress, intonation, emphasis)
4. Challenges in expressive speech generation
5. High computational and latency requirements
6. Poor support for low-resource languages and accents
7. Privacy and bias concerns
8. Difficulty handling long conversations and audio streams
9. Lack of standardized evaluation metrics

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Can we build our own LLM?

- ChatGPT (GPT-4o) ~200B
- Gemini 1.5 Pro 1T
- Claude 3 Opus ~200B
- Llama 3.1 405B 405B
- Mistral Large 2 123B
- Qwen2.5 72B 72B
- Perplexity AI 70B
- Grok 1.5 ~314B
- DeepSeek-V2 236B
- PaLM 2 ~540B
- DBRX Instruct 132B
- Microsoft Copilot ~20B

Each LLM is trained with billions of parameters!

Can we build our own LLM?

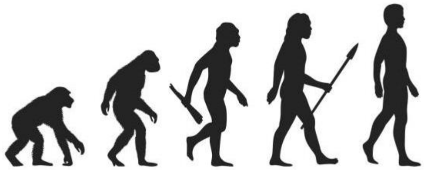
Everything should be made as simple as possible

- Albert Einstein

Example: The Transformer itself was built on a surprisingly simple idea, 'attention'. Sometimes the biggest breakthroughs come from better ideas, not bigger models

Outline

Evolution of Language Models

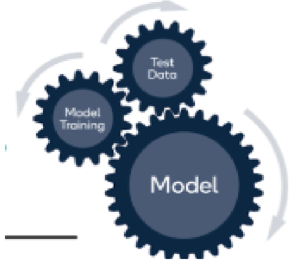


Speech LLMs

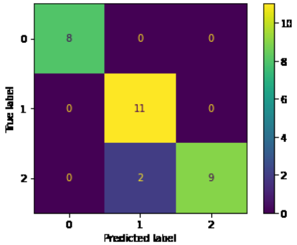
Various Types of Speech LLMs

Structure and Design choices of Speech LLMs

Training Speech LLMs

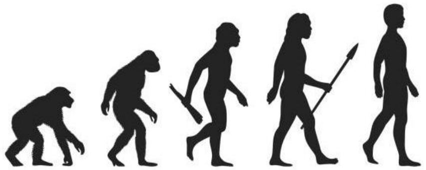


Evaluation



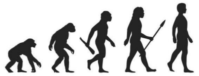
Outline

Evolution of
Language Models

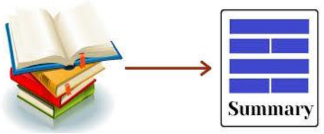


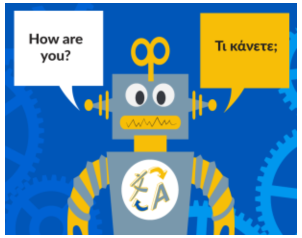
Text Domain

Evolution of LLMs

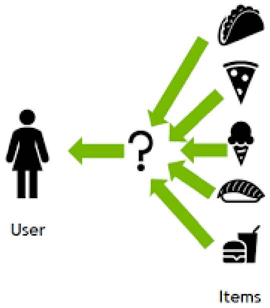


Task-specific

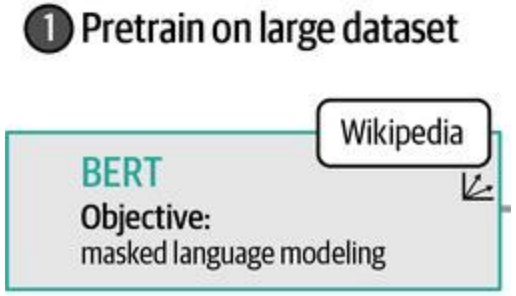




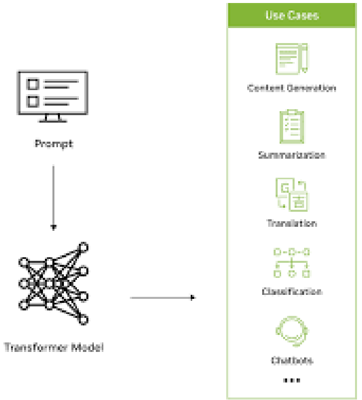

NER



Pre-trained representations



Generative Language Models



GPT, LLaMA etc.,

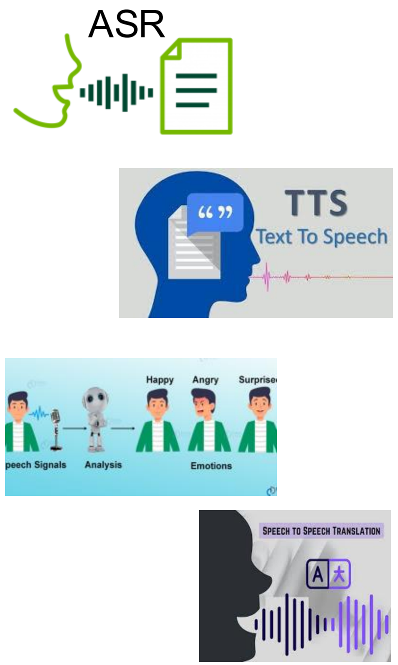
Instruction following & Chat systems

ChatGPT, LLaMA-Instruct etc.,

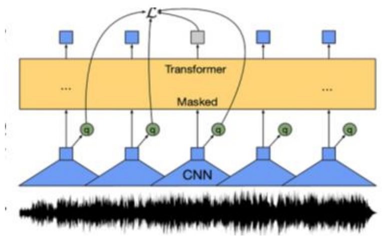
Speech Domain

Evolution of LLMs

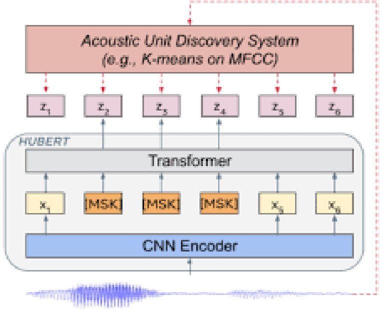
Task-specific



Pre-trained representations



wav2vec



HuBERT

Generative Language Models

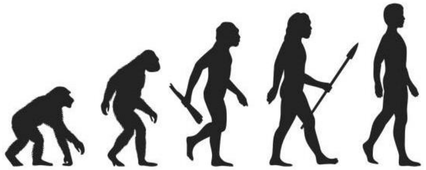
?

Instruction following & Chat systems

?

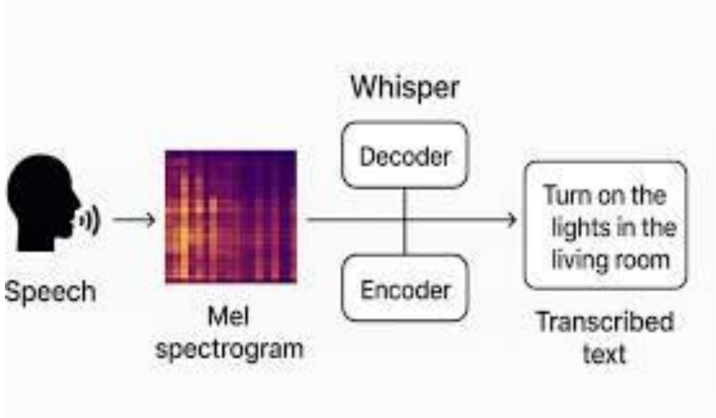
Outline

Evolution of
Language Models



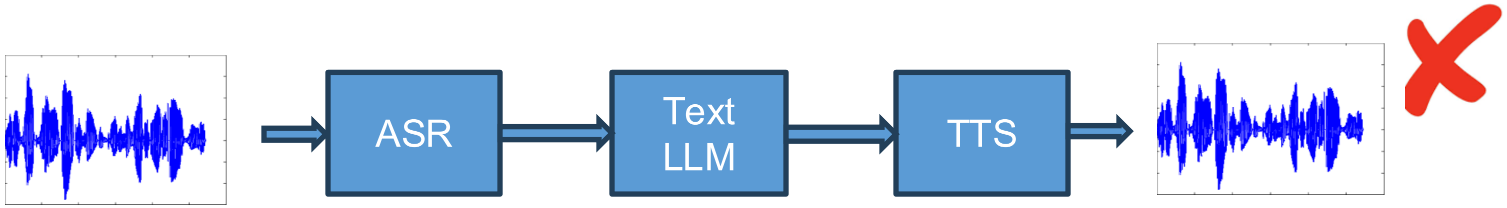
Speech LLMs

- Why ASR is not a Speech LLM?

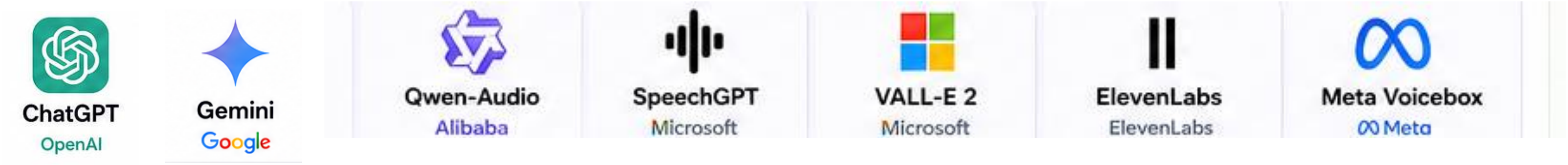


Trained with 680,000 hours of multilingual and multitask audio

Speech LLM



Speech / Voice agents:



Definition – Speech LLM

ASR

→ **Speech to Text**

→ **Text to Speech**

Well-defined formulations

Speech LLM → **?**

Input?

Output?

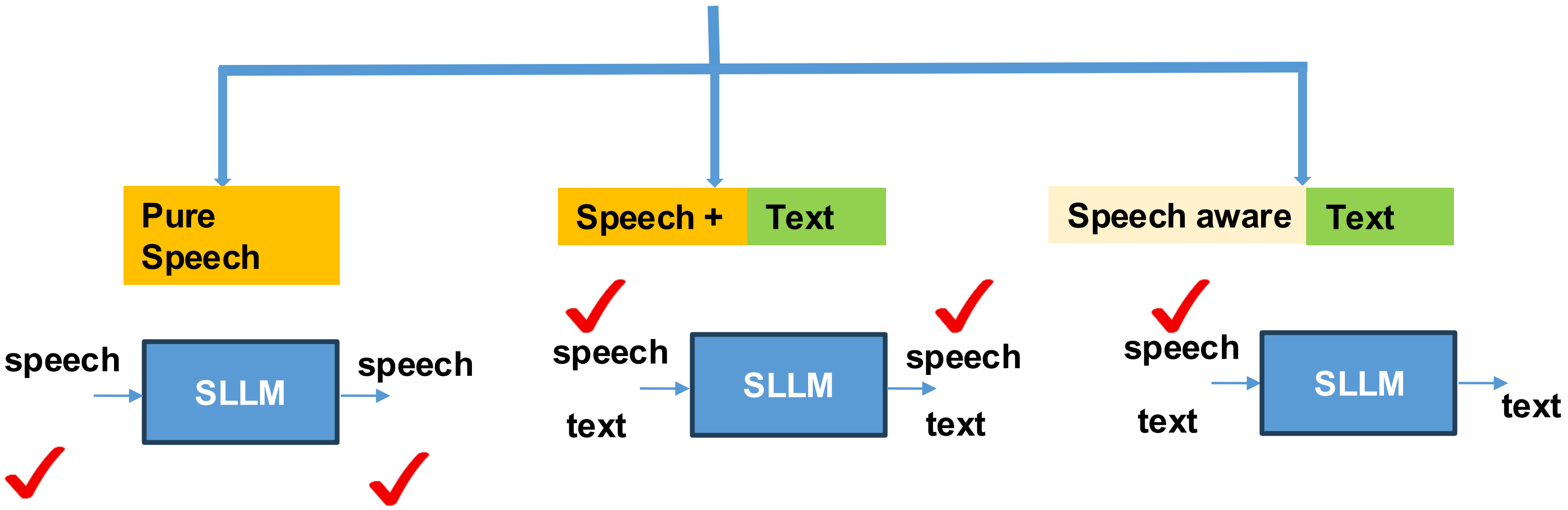
Definition – Speech LLM

Spoken Language Model ⇨ ?

Input?

Output?

Literature: No single definition

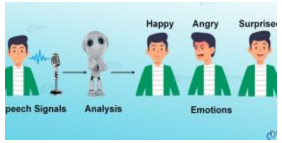


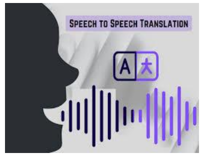
Definition – Speech LLM

Should satisfy three conditions:

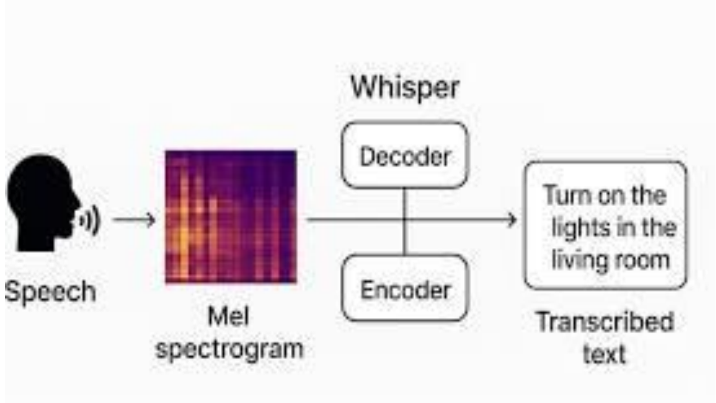
1. input and/or output must be speech ✓
2. be able to solve a variety of spoken language tasks ✓
3. takes instructions or prompts in the form of natural language ✓

"Write a short email to my manager requesting a day off next Friday"





- Why ASR is not a Speech LLM?



- Transcribe
- Translate
- Language Identification
- Voice Activity Detection

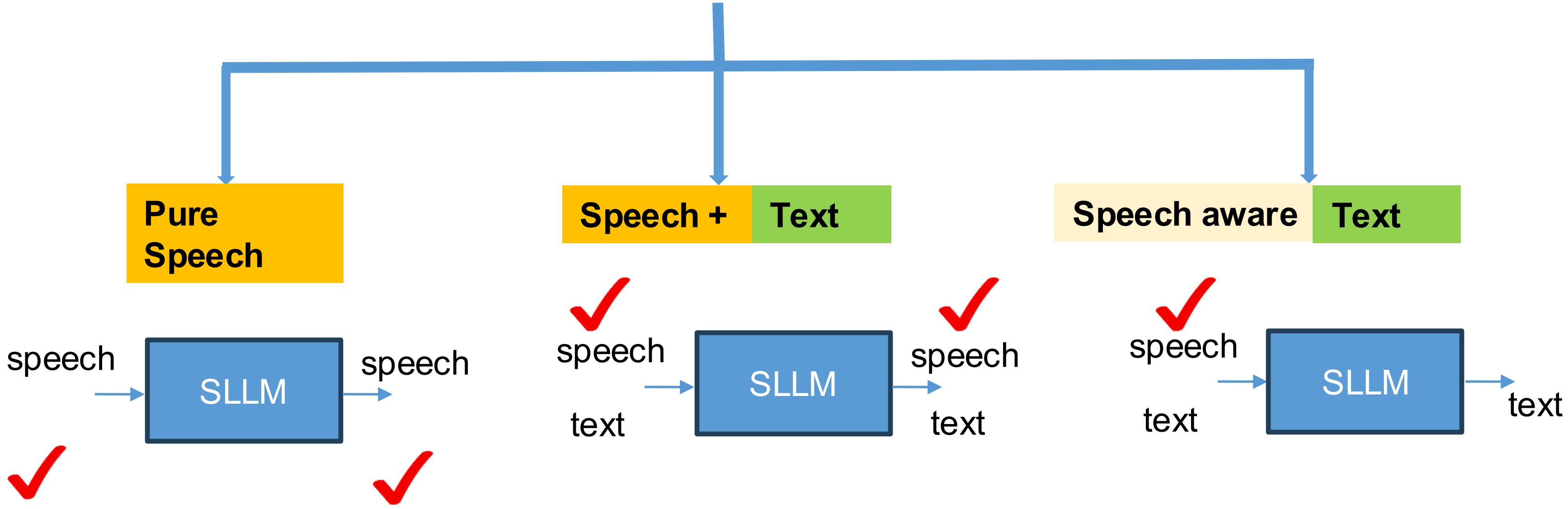
Trained with 680,000 hours of multilingual and multitask audio

Because, it can't take instructions/prompts in the form of natural language

Types of Speech LLMs

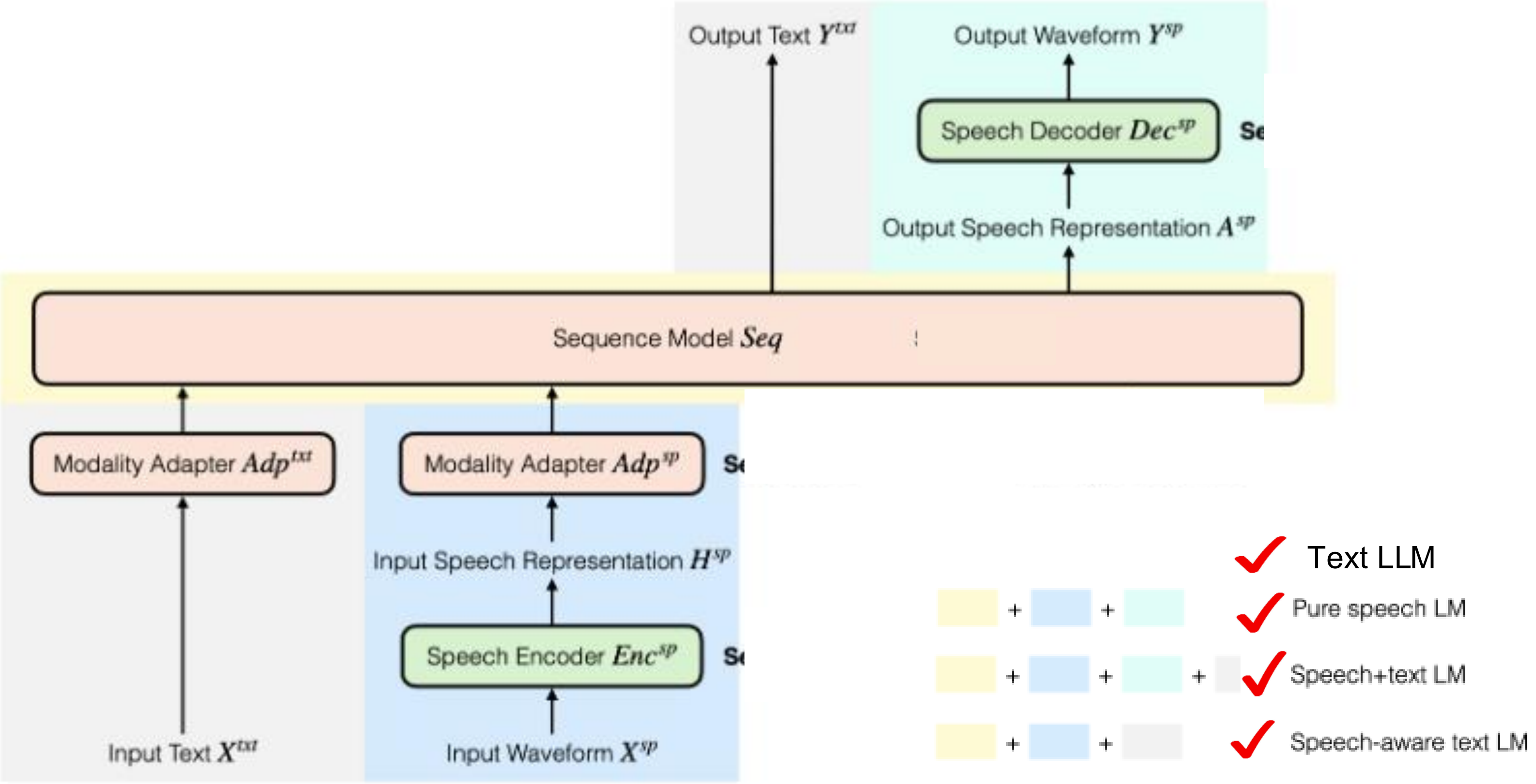
Speech LLMs
Types

All are valid Speech LLMs !



Design - Speech LLMs

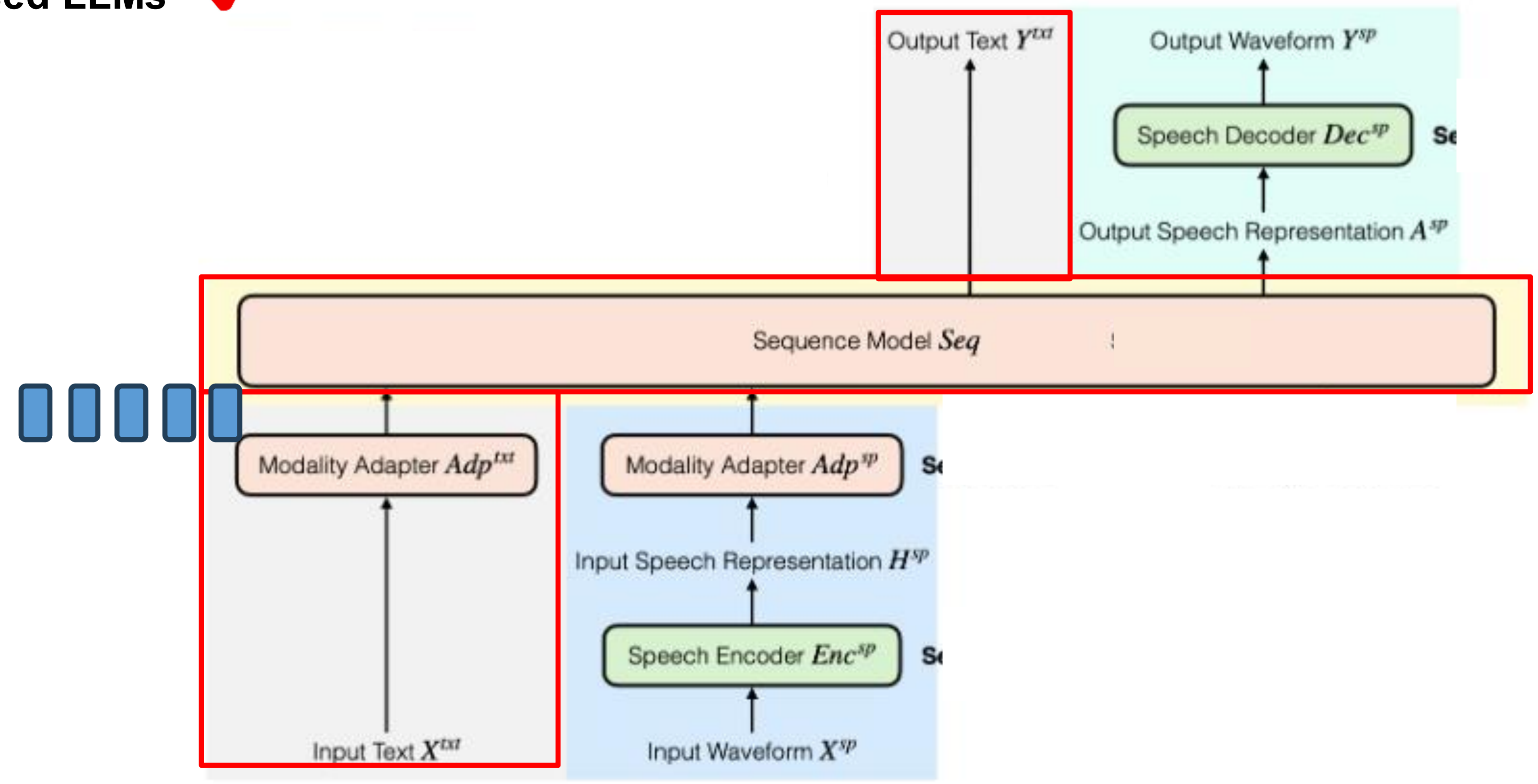
Design of Speech LLMs



Design - Speech LLMs

Design of Speech LLMs

Text-based LLMs ✓

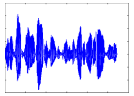


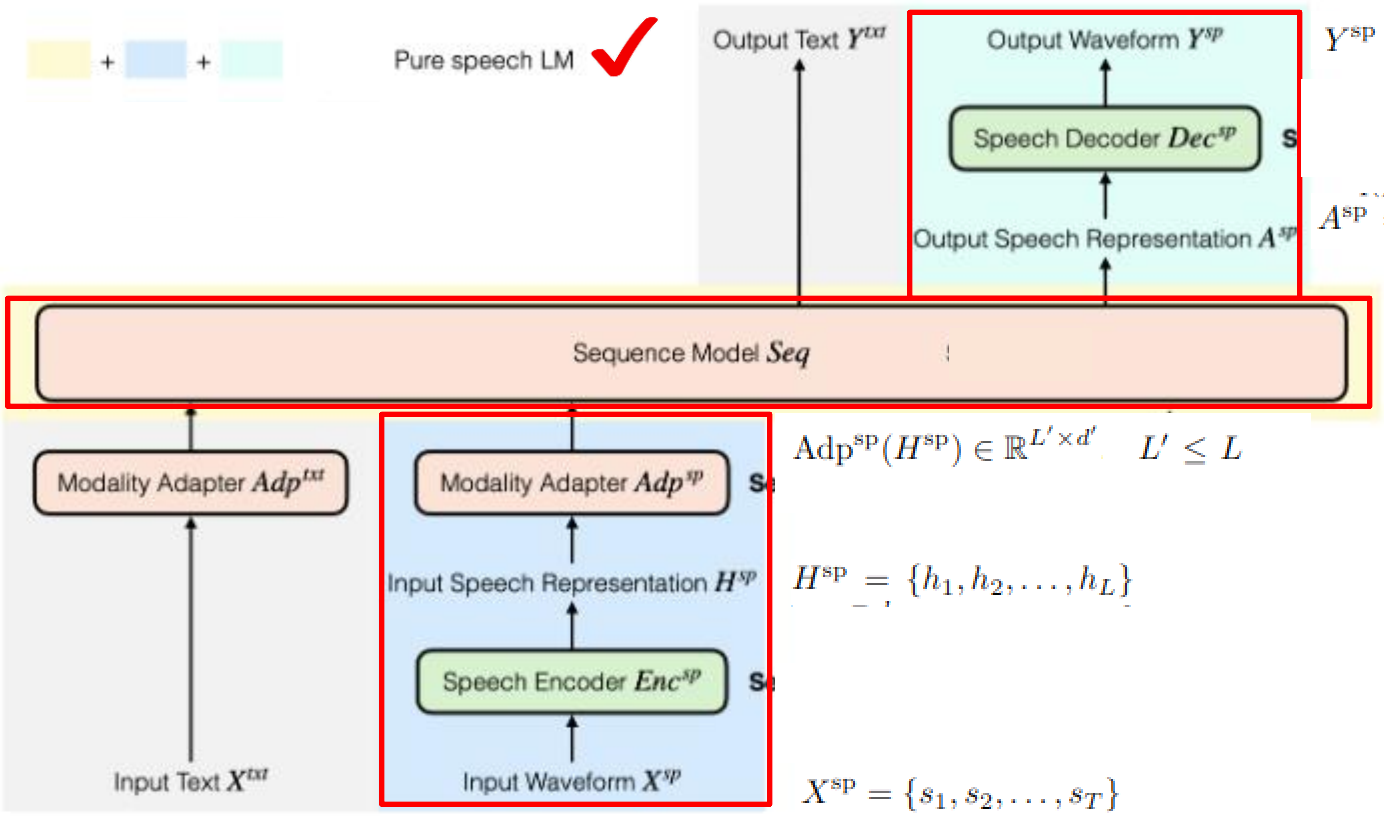
Autoregressive
(token by token generation)

This is a cat

Design - Speech LLMs

Design of Speech LLMs





$Y^{\mathrm{sp}} = \mathrm{Dec}^{\mathrm{sp}}(A^{\mathrm{sp}})$

$A^{\mathrm{sp}} = \{h'_1, h'_2, \ldots, h'_{L''}\}$

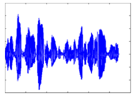
$h' = \mathrm{Seq}(\mathrm{Adp}^{\mathrm{sp}}(H^{\mathrm{sp}}))$.

autoregressive

$\mathrm{Adp}^{\mathrm{sp}}(H^{\mathrm{sp}}) \in \mathbb{R}^{L' \times d'}$ $\quad L' \leq L$

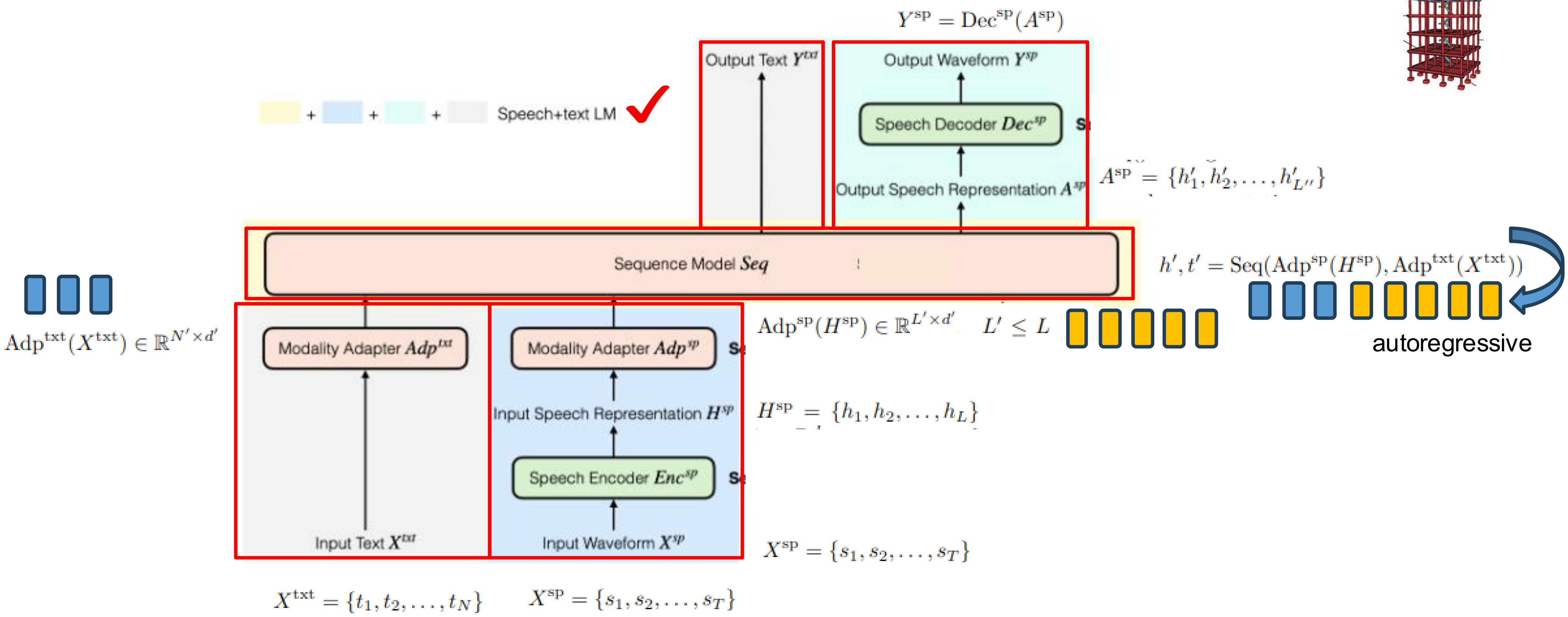
$H^{\mathrm{sp}} = \{h_1, h_2, \ldots, h_L\}$

$X^{\mathrm{sp}} = \{s_1, s_2, \ldots, s_T\}$



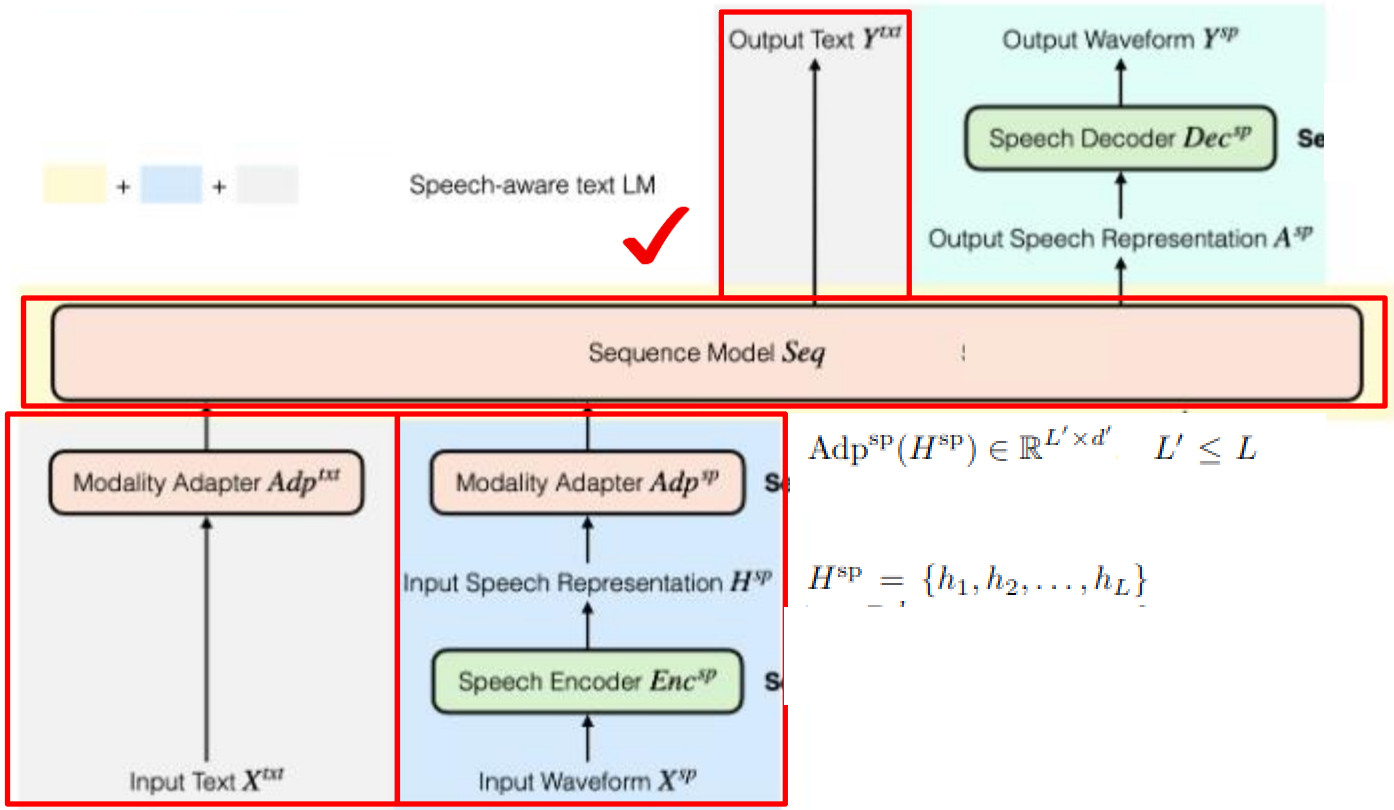
Design - Speech LLMs

Design of Speech LLMs



Design - Speech LLMs

Design of Speech LLMs



$$t' = \mathrm{Seq}\Big(\mathrm{Adp}^{\mathrm{sp}}(H^{\mathrm{sp}}),\ \mathrm{Adp}^{\mathrm{txt}}(X^{\mathrm{txt}})\Big)$$

autoregressive

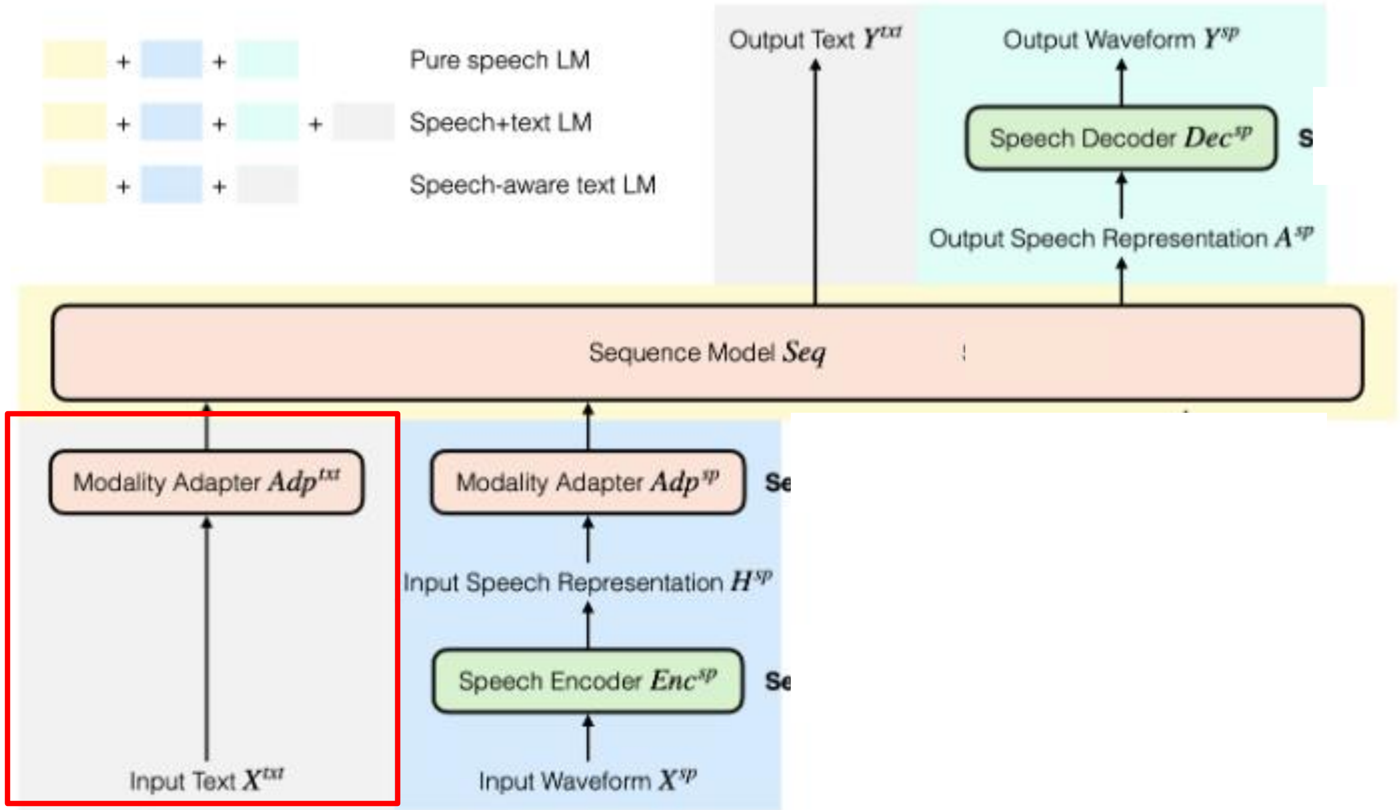
$$\mathrm{Adp}^{\mathrm{txt}}(X^{\mathrm{txt}}) \in \mathbb{R}^{N' \times d'}$$

$$\mathrm{Adp}^{\mathrm{sp}}(H^{\mathrm{sp}}) \in \mathbb{R}^{L' \times d'} \quad L' \leq L$$

$$H^{\mathrm{sp}} = \{h_1, h_2, \ldots, h_L\}$$

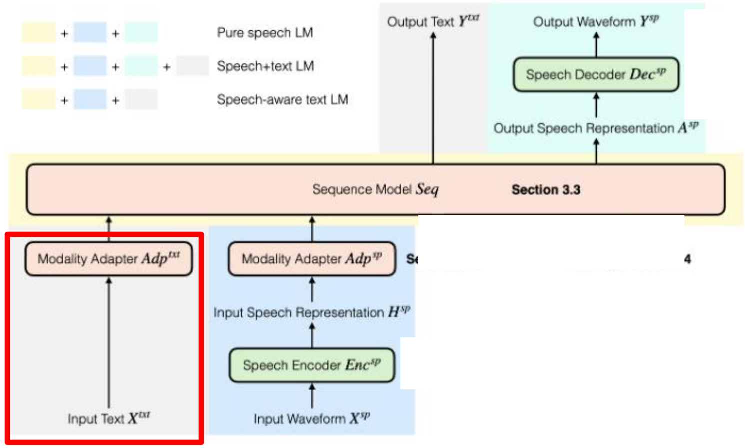
$$X^{\mathrm{txt}} = \{t_1, t_2, \ldots, t_N\} \quad X^{\mathrm{sp}} = \{s_1, s_2, \ldots, s_T\}$$

Design - Speech LLMs



Modality Adapter

Design of Speech LLMs



Objective:

To learn better representations

Representations?

Apple released a phone.

Apple is rich in fibre.

Modality Adapter

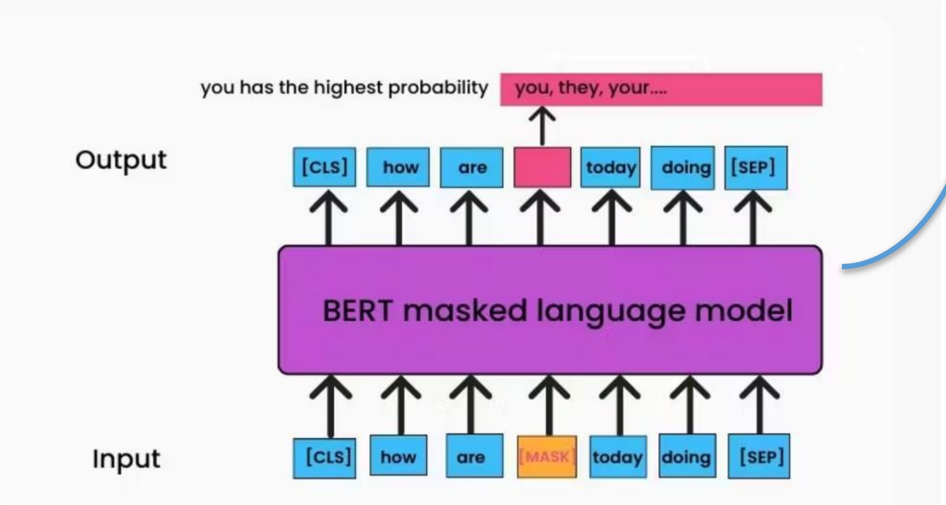
Design of Speech LLMs

BERT

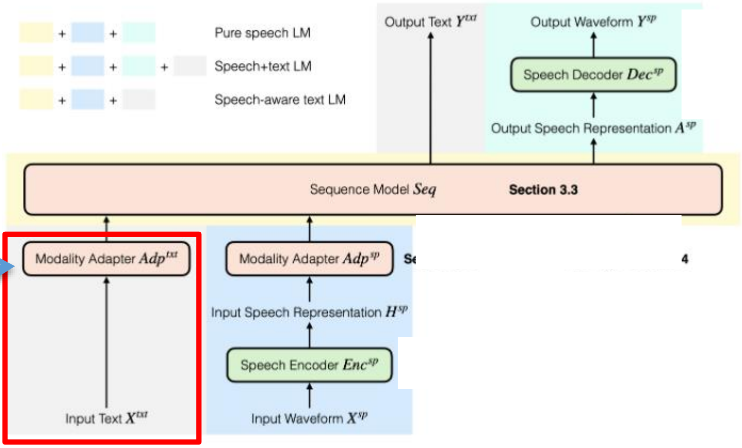
Objective:

masked language modeling

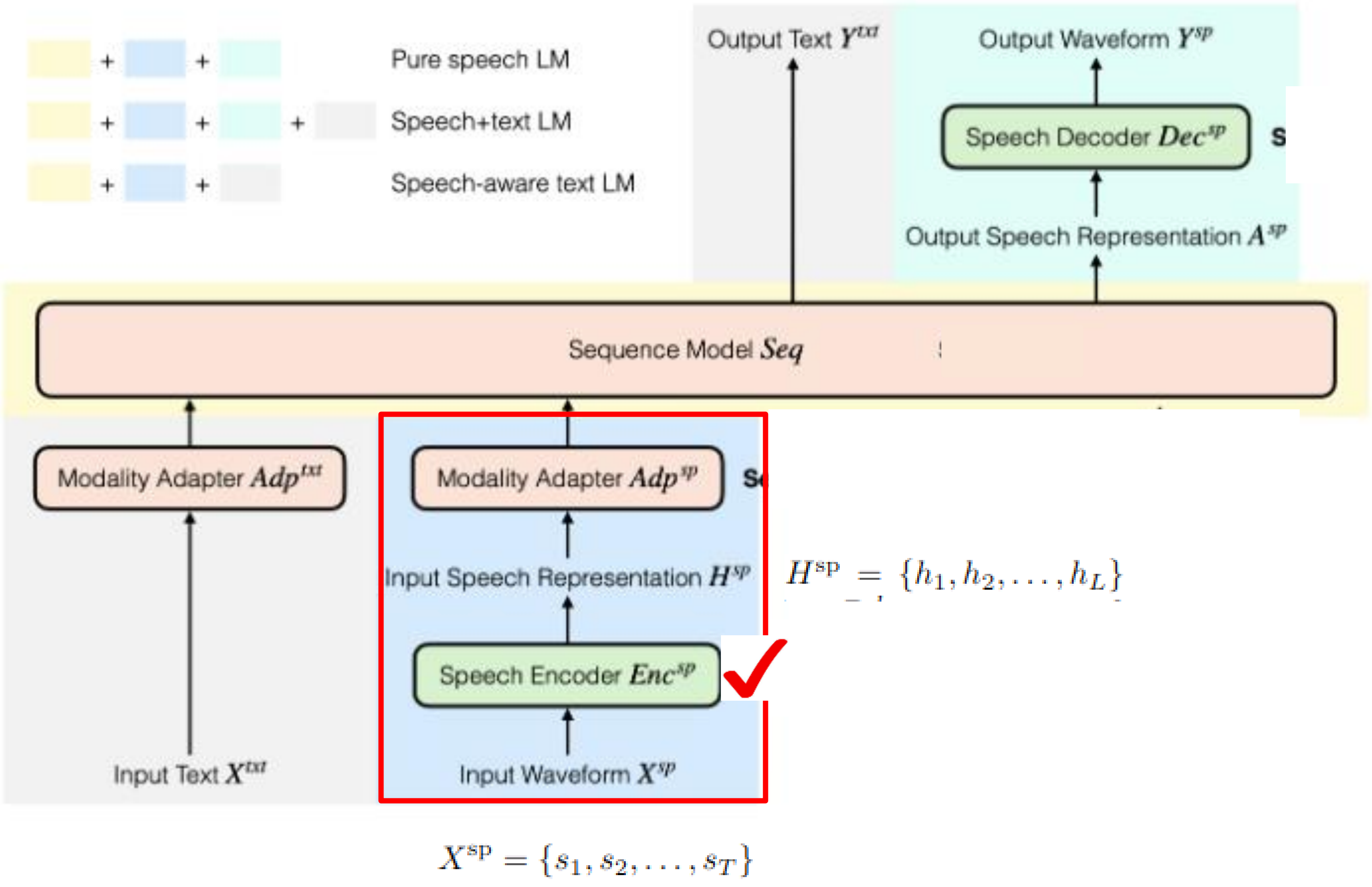
Wikipedia



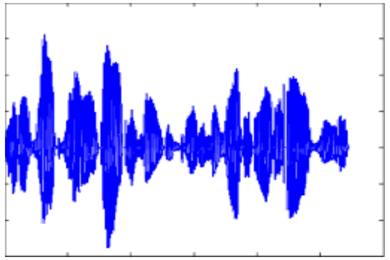
Training



Design - Speech LLMs



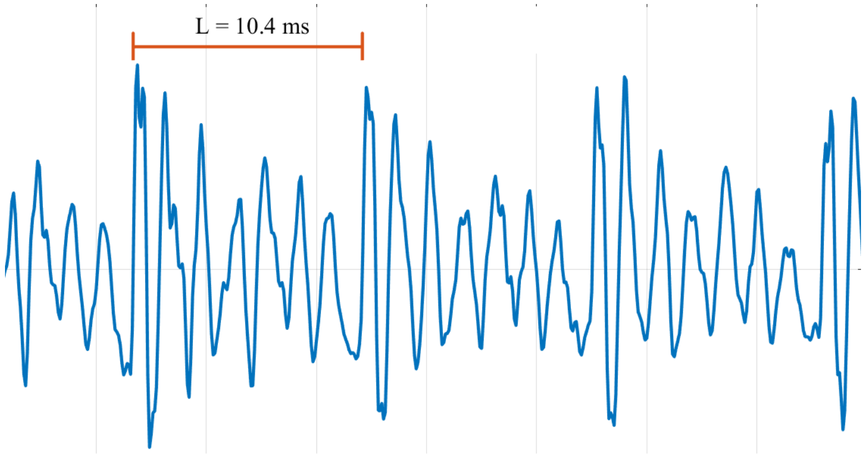
Need of Speech Encoder



1 sec

16000 samples (16 kHz)

1. **Continuous**
2. **Time-varying in nature**
3. **Non stationary**



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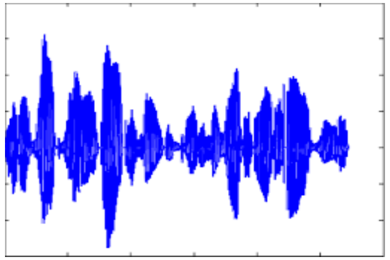
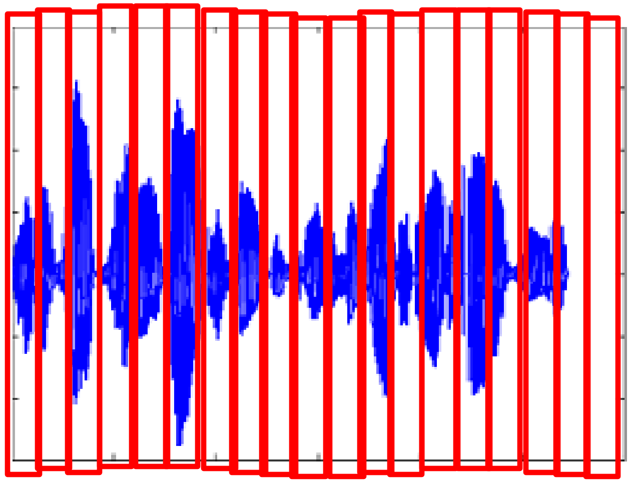


1 sec



Frames

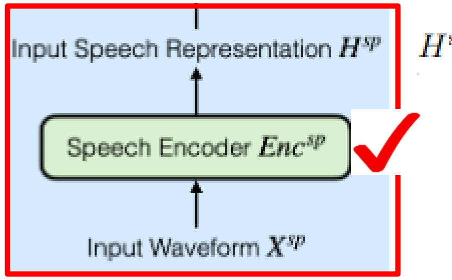
Each frame – 20 ms
Overlap – 10 ms

1. **Acoustic information**
2. **Phonetic information**
3. **Speaker information**
4. **Contextual information**

Speech Encoder

Discrete tokens:

1. **Phonetic tokens**
   - SSL -> clusters
2. **Audio codec tokens**
   - Vector Quantization (VQ)
   - Residual Vector Quantization (RVQ)
3. **Temporal compression**
   - Deduplication



$H^{sp} = \{h_1, h_2, \ldots, h_L\}$

$X^{sp} = \{s_1, s_2, \ldots, s_T\}$

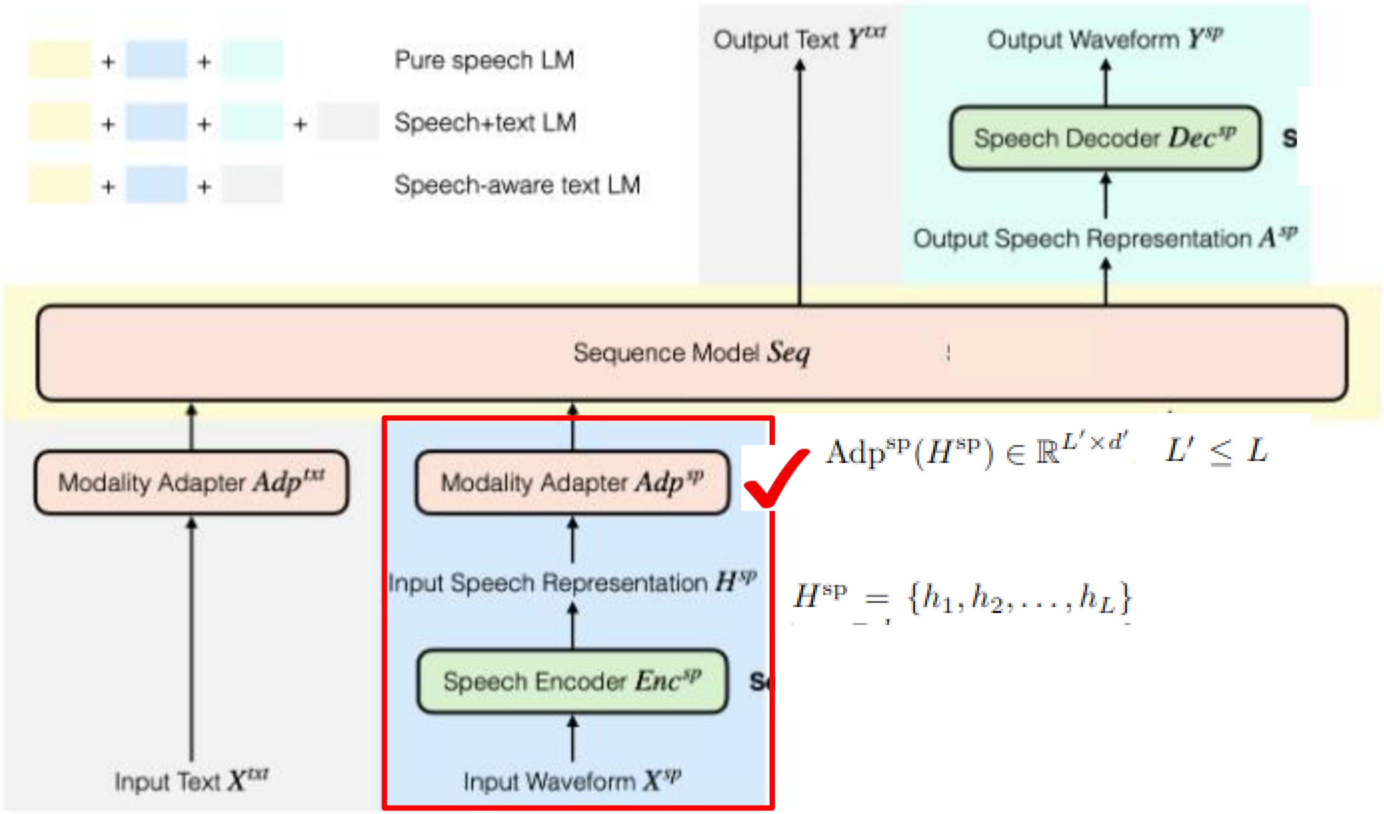
To get frame-level representations

Continuous features:

1. **Signal processing based**
   - MFCCs
2. **SSL features**
   - Wav2vec2
   - HuBERT
   - WavLM
   - XeUS
3. **Supervised pre-trained**
   - Whisper
   - USM
4. **Neural audio codecs**
   - SoundStream
   - EnCodec

Modality Adapter

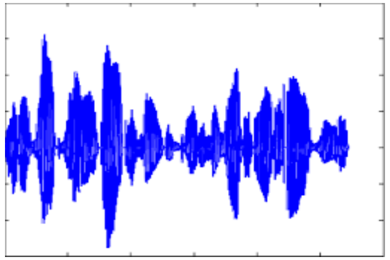
To compress the features



$\mathrm{Adp}^{\mathrm{sp}}(H^{\mathrm{sp}}) \in \mathbb{R}^{L' \times d'}$ $\quad L' \leq L$

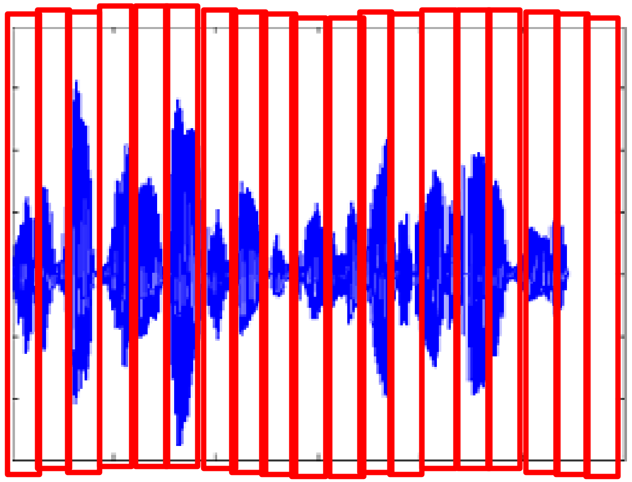
$H^{\mathrm{sp}} = \{h_1, h_2, \ldots, h_L\}$

Need of Modality Adapter



1 sec

160 Frames



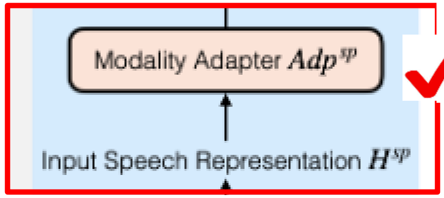
Frames

Each frame – 20 ms
Overlap – 10 ms

10 sec audio -> 1600 frames

10 sec audio -> Less than 100 words

Modality Adapter



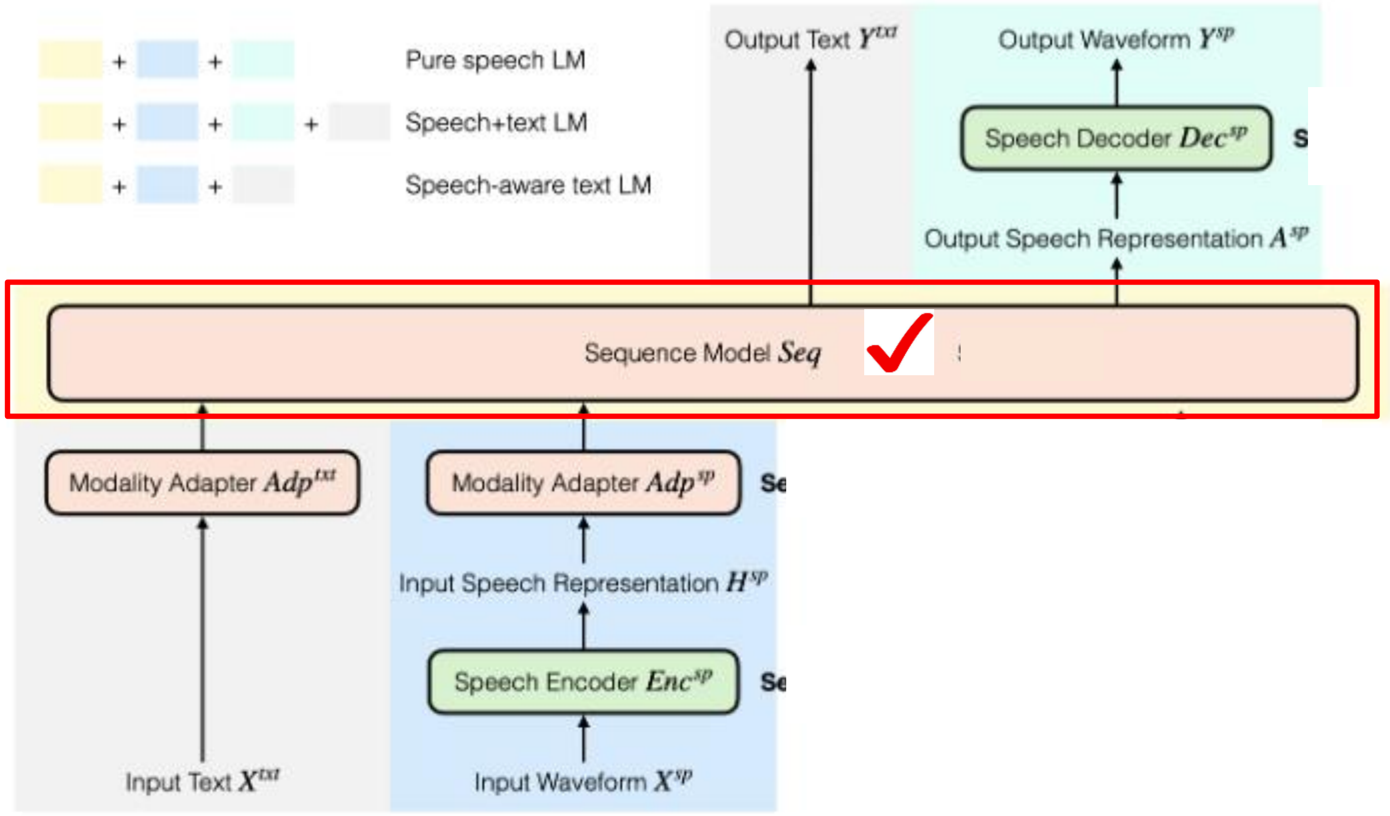
$\mathrm{Adp}^{\mathrm{sp}}(H^{\mathrm{sp}}) \in \mathbb{R}^{L' \times d'} \quad L' \leq L$

$H^{\mathrm{sp}} = \{h_1, h_2, \ldots, h_L\}$

To compress the features in time resolution

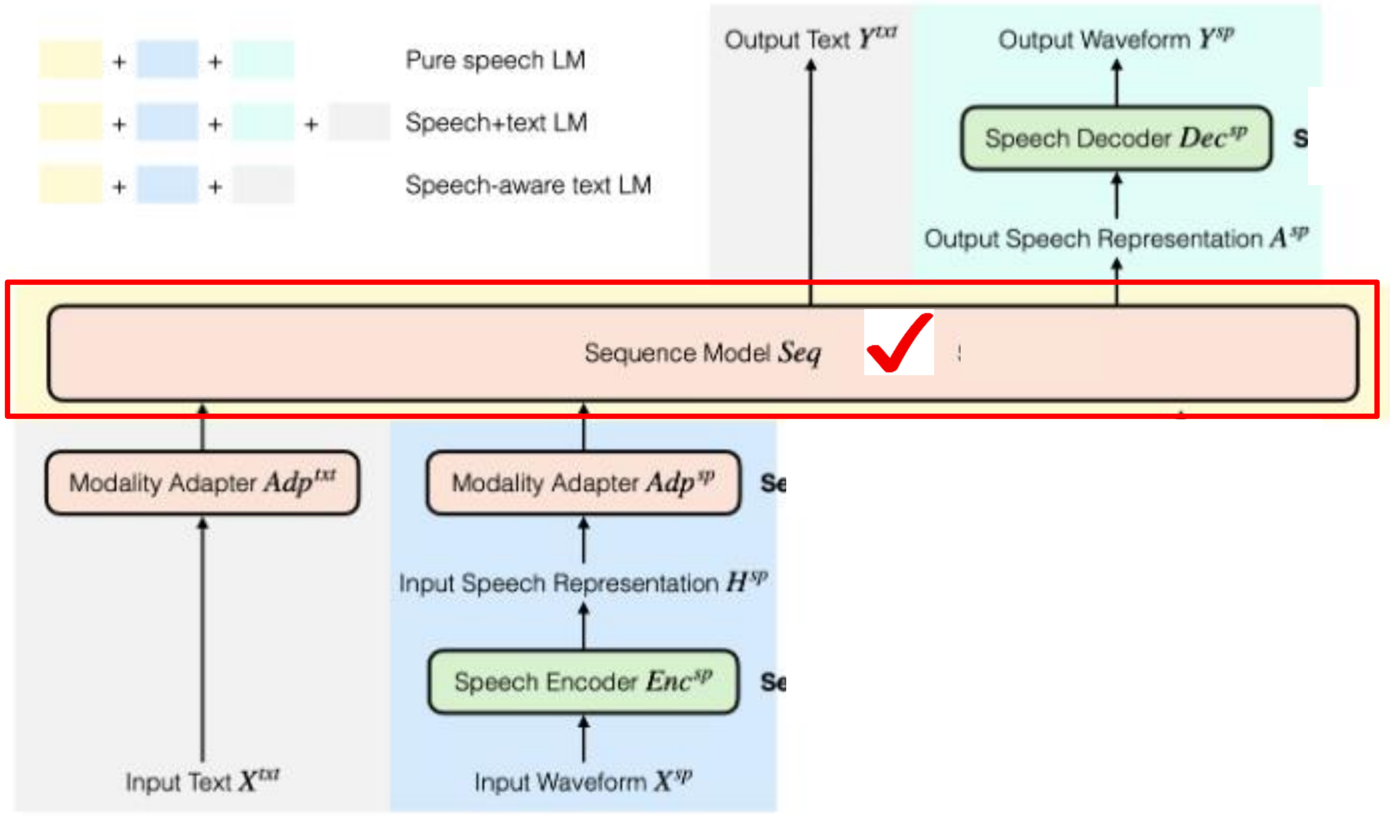
1. **CNN with Strides**
   - 500 frames → CNN → 100 frames
2. **CTC-based Compression**
   - Multiple phonetically related frames to single token
3. **Q-Former**
   - To achieve fixed-length of representation
   - (any length audio to fixed length vectors)

Design - Speech LLMs



Generating tokens (text and/or speech)

Design - Speech LLMs



Generating tokens (text and/or speech)

1. Hierachical token generation (speech)
2. Text and speech hybrid generation

Sequential Model

1. Hierachical token generation

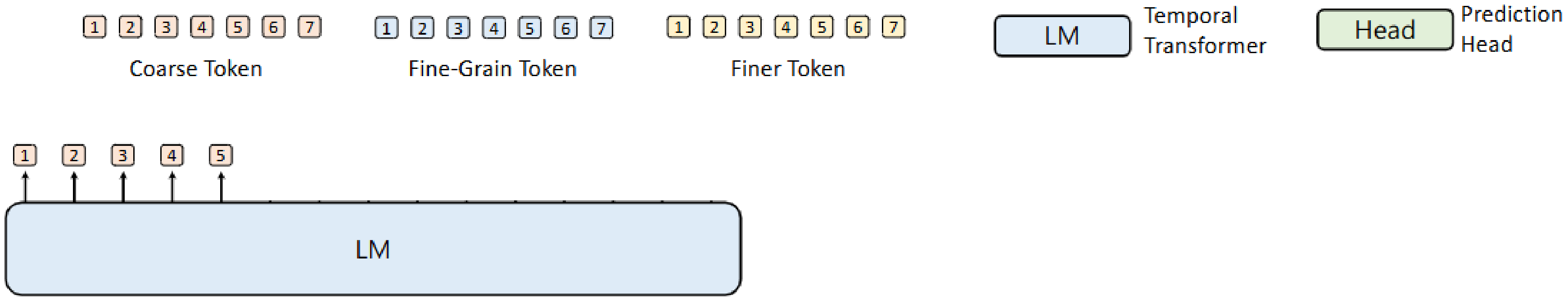
1 2 3 4 5 6 7
Coarse Token → High-level linguistic / semantic information (phoneme info)

1 2 3 4 5 6 7
Fine-Grain Token → Pronunciation, stress, prosody, speaker/style information

1 2 3 4 5 6 7
Finer Token → Waveform-level characteristics, harmonics, timbre, micro-prosody

Sequential Model

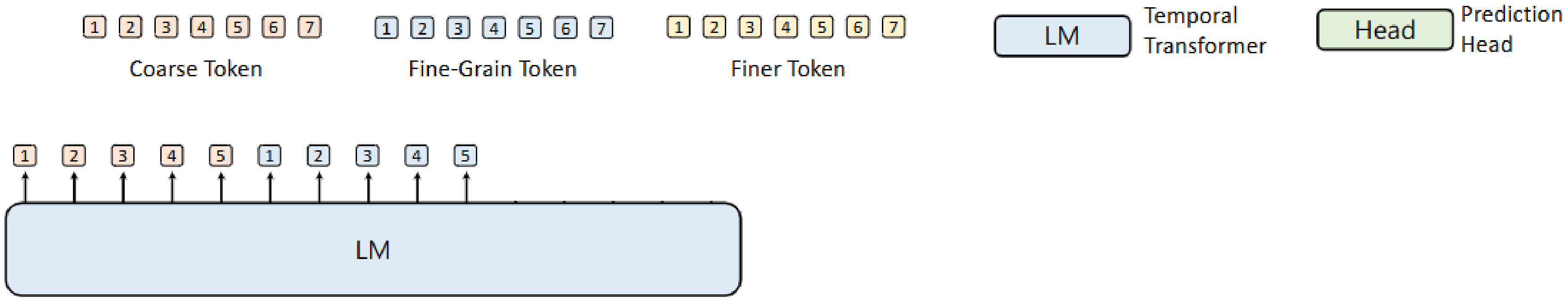
1. Hierachical token generation



(a) Coarse first, then fine-grained

Sequential Model

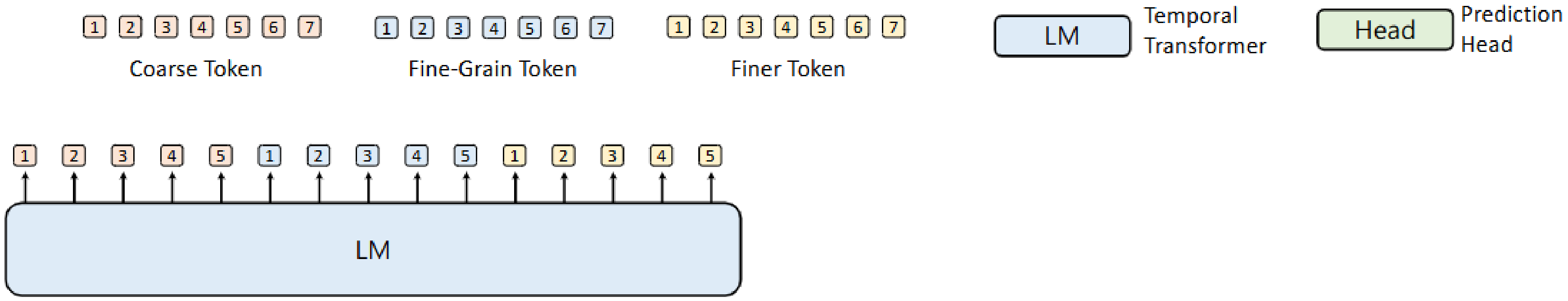
1. Hierachical token generation



(a) Coarse first, then fine-grained

Sequential Model

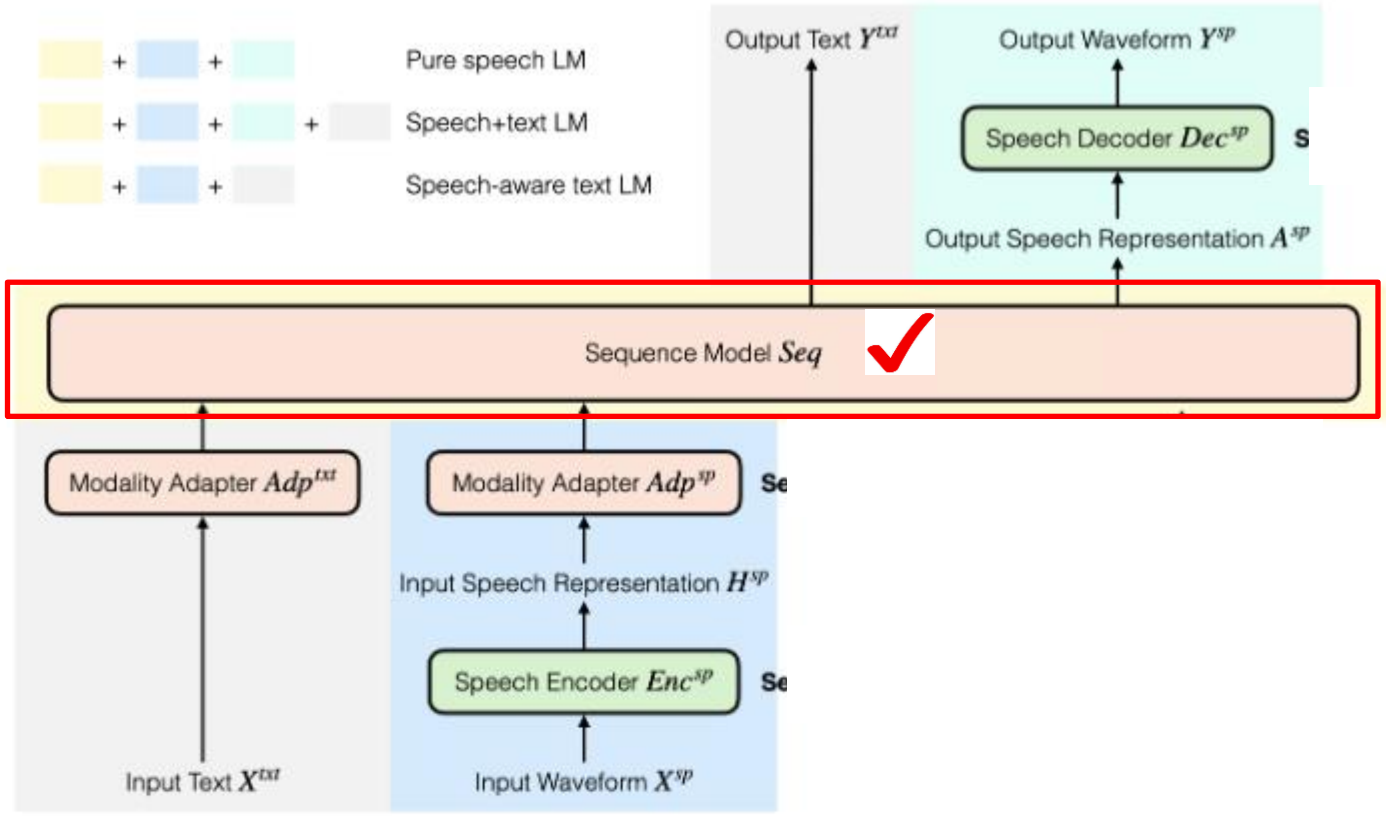
1. Hierachical token generation



(a) Coarse first, then fine-grained

Design - Speech LLMs

Design of Speech LLMs



Generating tokens (text and/or speech)

1. Hierachical token generation (speech)

2. Text and speech hybrid generation

Sequential Model

1. **Text and speech hybrid generation**

Mini-Omni

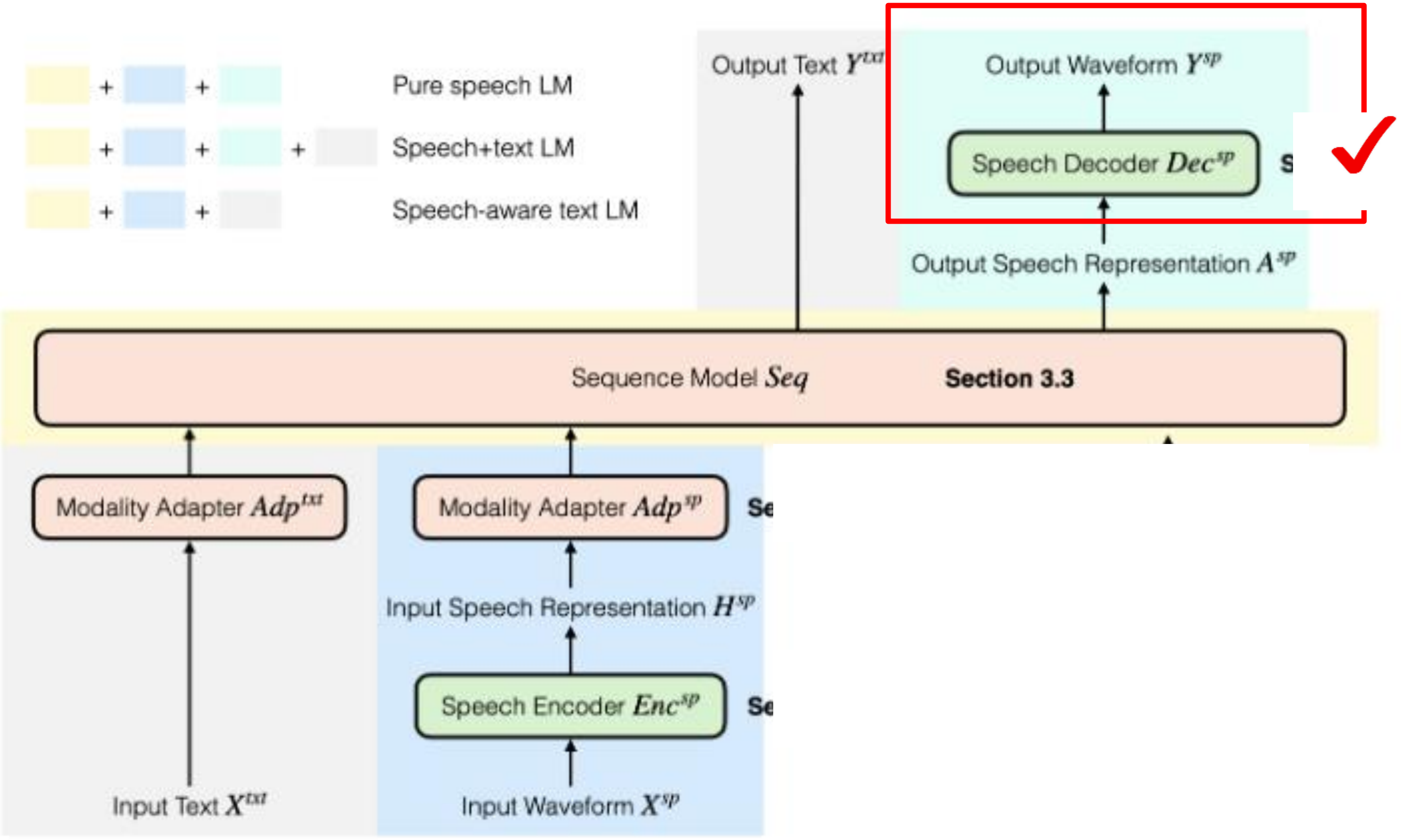
3	77	23	12	71	34	3	23
how	are	you	ε	ε	ε	ε	ε

SpiRit-LM

how	12	71	you	16	3	88	15

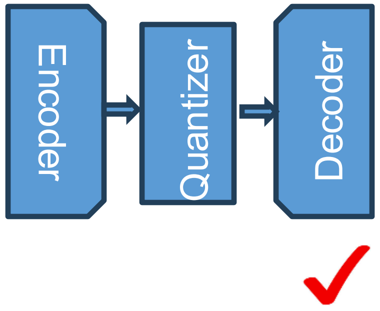
Speech Decoder

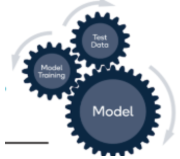
Design of Speech LLMs



Speech representations -> waveform

- Vocoder (world-vocoder)
- Neural-based vocoder (Hifi-GAN)
- Audio-codec decoder





Training Strategies - Speech LLMs

Text-based LLMs:

Sequence Model *Seq*

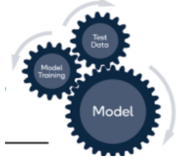
1. **Pre-training:** Next token prediction

Input (context)	Target (next token)
The	cat
The cat	sat
The cat sat	on
The cat sat on	the
The cat sat on the	mat

2. **Post-training:** Instruction-based

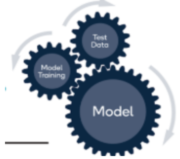
Instruction: Translate to French
Input: Hello
Output: Bonjour

Training Strategies - Speech LLMs



Speech LLMs:

1. **Pre-training:** Next token prediction
   - Pure speech pre-training p(speech)
   - Joint speech and text pre-training p(text, speech)
   - Conditional pre-training p(text|speech)



Training Strategies - Speech LLMs

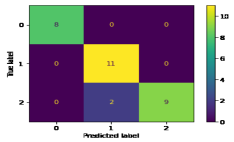
Speech LLMs:

1. **Post-training:**

Task	Example Instructions
Speech Recognition (ASR)	Recognize the speech and provide the transcription Repeat the spoken content in English
Speech Translation	Translate the given speech into English Recognize the speech and then translate it
Speaker Recognition	How many speakers are present in the audio? Identify who the speakers are
Emotion Recognition	Describe the emotion of the speaker Classify the emotion (sad, angry, neutral, happy)
Question Answering (Audio-based)	What happened to the person in the audio? Generate a factual answer to the question What medicine is mentioned? Briefly explain it

Evaluation – Speech LLMs

Evaluation

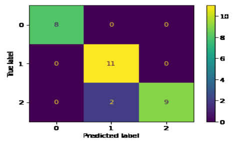


Traditional methods:

- ASR-based
  - WER
- TTS-based
  - Speech naturalness
  - Intelligibility
  - Speaker/style consistency
  - Response relevance and correctness
- Speech Translation-based
  - BLEU score

Evaluation – Speech LLMs

Evaluation

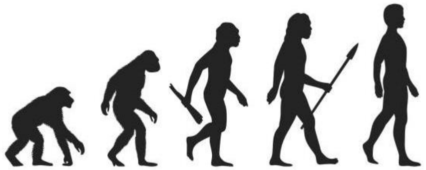


Advanced methods:

- Trustworthiness
    - Hallucinations
    - Toxic responses
    - Bias across speakers
    - Deepfake vulnerability
- Interactivity
    - Response latency
    - Turn-taking
    - Interruption handling

Conclusion

Evolution of
Language Models

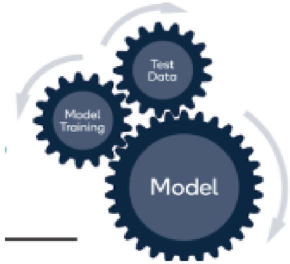


Spoken Language
Modeling (SLM)

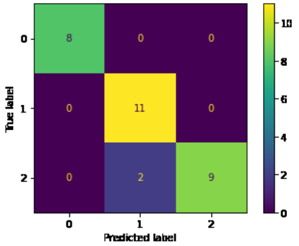
Various Types
of SLMs

Structure and Design
choices of SLMs

Training SLMs



Evaluation



Thank You!

Questions?